

Cell Membrane and Its Functions

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02_Cell_membrane_and_its_functions.pdf (25 slides) · Refs: Guyton & Hall 13th ed.; Ganong 25th ed.

Legend: ★ = high-yield · [added] = not on slides ·  = the lecturer's handwritten annotation on the slide (these signal emphasis). Images marked "Slide N" come from the lecture.

 **Bridge from Lecture 01 (Intro to Physiology):** Lecture 01 showed that ICF (K⁺-rich) and ECF (Na⁺-rich) differ sharply, while osmolarity stays at 290 everywhere. This lecture explains **the barrier that keeps them different** (the cell membrane) and starts on **how substances cross it**.

1. Title / Objectives

Objective	Likely tested as
★ Composition of the cell membrane	Percentages (55/25/13/4/3); amphipathic phospholipid
★ Functions of the membrane (proteins)	Integral vs peripheral; junction types
★ Methods of transport	Classification tree; simple vs facilitated diffusion; diffusion formula

Flow: Structure & composition → functions of membrane proteins → transport classification → simple diffusion (bilayer, channels) → facilitated diffusion → conclusion → case.

2. Main Content (Lecture Order)

2.1 Structure & Composition (Slides 4-7)

- Surrounds the cell completely; **very thin, elastic, semipermeable**
- ★ **Thickness 7.5-10 nm** (= **75-100 Å**; 1 Å = 10⁻¹⁰ m)

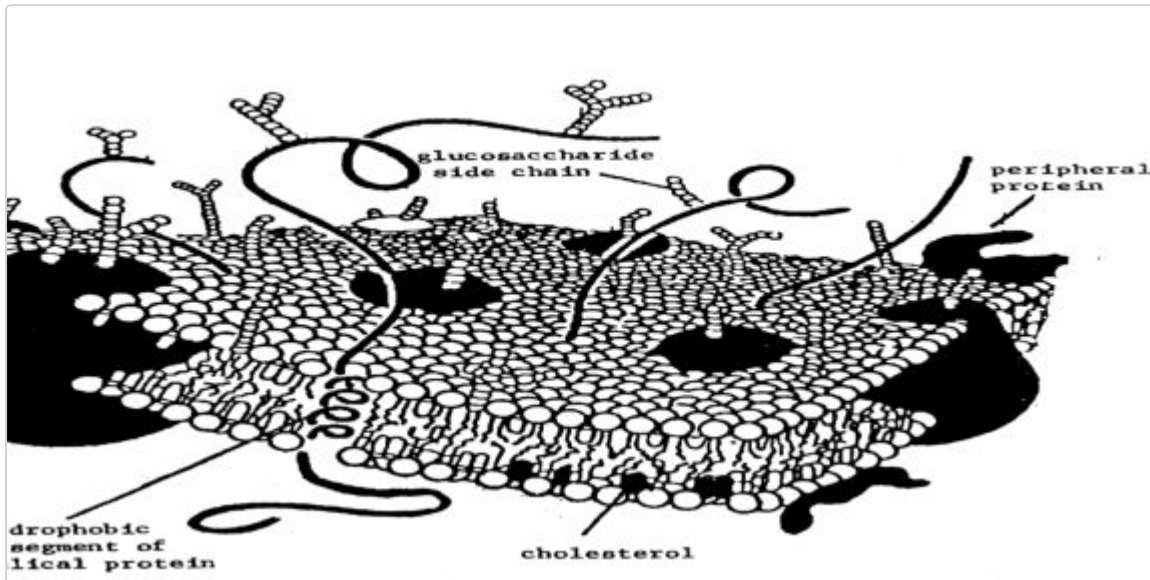
★★ **Composition:**

Component	%
★ Proteins	★ 55%
Phospholipids	25%
Cholesterol	13%
Other lipids	4%
Carbohydrates	3%
<i>Mnemonic [added]: "55-25-13-4-3" (proteins > phospholipids > cholesterol > other lipids > carbohydrates).</i>	

★ **Phospholipid = "clothespin" shape:**

Part	Chemistry	Faces
Head	Phosphate radical: water-soluble, hydrophilic, polar	The aqueous ECF outside and the cytoplasm inside
Two tails	Fatty acids: insoluble in water, hydrophobic, non-polar	Each other, in the water-poor membrane interior

★ So the molecule (and the membrane) is **amphipathic**: one part hydrophilic, the other hydrophobic.



Slide 5: Fluid mosaic membrane: bilayer, integral and peripheral proteins, cholesterol, glycocalyx

Role of each component (Slide 7):

1. **Lipid bilayer: flexibility and selective permeability**
2. ★ **Cholesterol**: affects **permeability** and gives **toughness**
3. **Proteins** float in the bilayer: **integral** and **peripheral** types
4. ★ **Carbohydrates** occur combined as **glycoproteins and glycolipids** (the glycocalyx [term added]). They act as:
 1. **Recognition sites**
 2. **Self-differentiation**
 3. **Immune responses**
 4. **Attaching cells together**

🔗 **Recap: Structure.** 7.5–10 nm, semipermeable. 55% protein, 25% phospholipid, 13% cholesterol, 4% other lipids, 3% carbohydrate. Amphipathic phospholipids form the bilayer (flexibility, selectivity). Cholesterol adds toughness. Carbohydrates handle recognition and immunity.

🔗 **Link:** Proteins are the largest share (55%) and do most of the membrane's work.

▶ **Up next: Functions of membrane proteins.**

2.2 Functions of Cell Membrane Proteins (Slides 8–10)

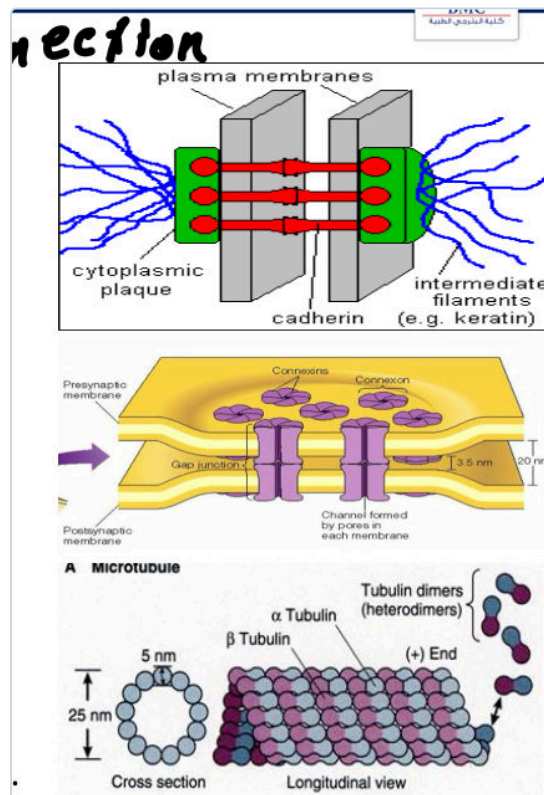
#	Function	Detail	📝 Lecturer's grouping
1	Structural	Keep membrane integrity and strength	

#	Function	Detail	Lecturer's grouping
2	Passive channels		Integral
3	Carriers in facilitated diffusion		Integral
4	Carriers in active transport (pumps)		Integral
5	Receptors		Peripheral
6	Enzymes		Peripheral
7	★ Identity proteins	Mostly glycoproteins ; give the cell the individual's identity label so the immune system doesn't attack it	
8	★ Intercellular connections	See below	"8 and 9: connection"
9	Cell adhesion molecules (CAMs)	Attach the cell to the basal lamina	connection
10	Cytoskeleton	Fibres that maintain shape and let the cell change shape and move	

[added caveat: receptors are often integral (transmembrane) proteins in textbooks; the lecturer's handwriting groups them with peripheral, so follow the lecture for this exam.]

★★ Intercellular connections (Slide 10):

Purpose	Junctions
Fasten cells to each other and to surrounding tissue	1. ★ Tight junctions (zonula occludens) · 2. Desmosomes · 3. Zonula adherens
Transfer ions and molecules between cells	★ Gap junctions (connexons, ~3.5 nm gap on the figure)



Slide 10: Desmosome, gap junction and microtubule (cytoskeleton)

🔗 **Recap: Protein functions.** 10 functions. Channels and carriers/pumps are integral; receptors and enzymes are peripheral (per the lecturer). Identity glycoproteins protect against the immune system. Tight junctions, desmosomes and zonula adherens hold cells together; gap junctions let ions pass between them. CAMs bind to the basal lamina; the cytoskeleton gives shape and movement.

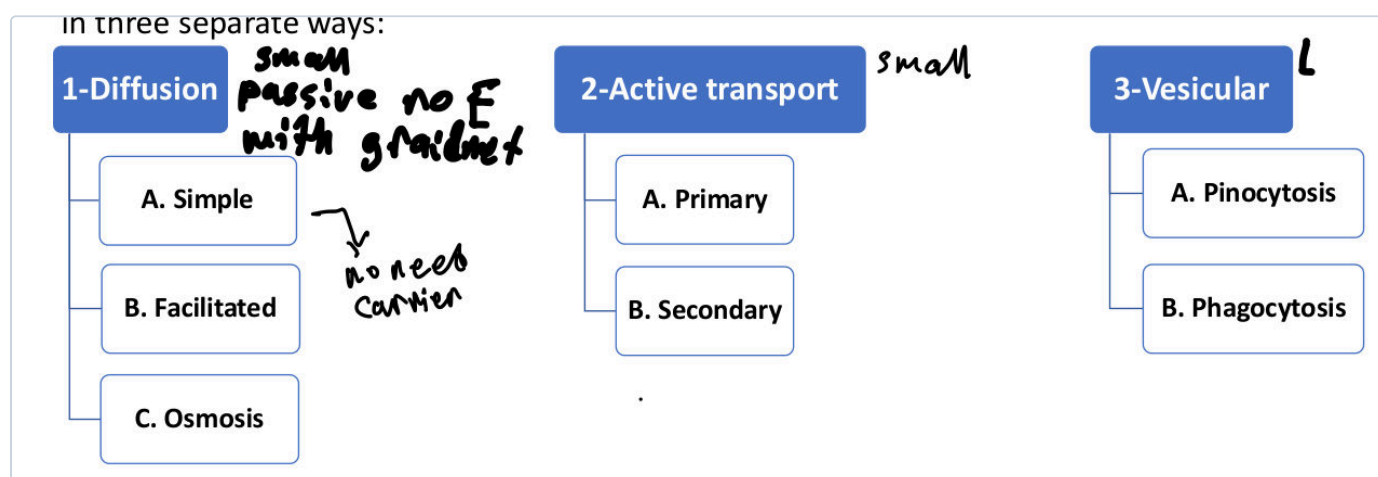
🔗 **Link:** Three of those protein functions (channels, facilitated carriers, pumps) are transport. The rest of the lecture classifies how substances cross.

🔗 **Up next: Methods of transport.**

2.3 Methods of Transport: Classification (Slides 11-12)

The ICF-ECF differences are vital to the cell; transport mechanisms maintain them. **Three ways:**

1. Diffusion	2. Active transport	3. Vesicular
A. Simple · B. Facilitated · C. Osmosis	A. Primary · B. Secondary	A. Pinocytosis · B. Phagocytosis
📝 Small; passive, no energy, with the gradient	📝 Small	📝 Large (L)
📝 Simple = no carrier needed		



Slide 12: Transport classification with the lecturer's notes

🔗 **Recap: Classification.** Diffusion (simple, facilitated, osmosis): passive, small molecules. Active (primary, secondary): small molecules against the gradient. Vesicular (pino-, phagocytosis): large molecules.

🔗 **Link:** The lecture starts with the simplest route, diffusion with no carrier at all.

🔗 **Up next: Simple diffusion.**

2.4 Simple Diffusion (Slides 13-17)

★ **Definition:** molecules/ions diffuse through the membrane **along the concentration gradient, passively, without binding to carrier proteins.**

★★★★ **Rate of diffusion (Slide 13, starred by the lecturer):**

Diffusion rate $\propto$ (Concentration difference $\times$ Surface area $\times$ Temperature) $\div$ (Distance $\times$ $\sqrt{\text{Molecular weight}}$)

✎ Numerator = **direct** relation; denominator = **inverse** relation.

$$\text{Diffusion rate} \propto \frac{\text{Conc. Difference} \times \text{Surface area of the membrane} \times \text{Temp.}}{\text{Distance} \times \sqrt{\text{Molecular Weight}}}$$

Handwritten notes: "direct" with an arrow pointing to the numerator, and "inverse" with an arrow pointing to the denominator. A star is in the top right corner.

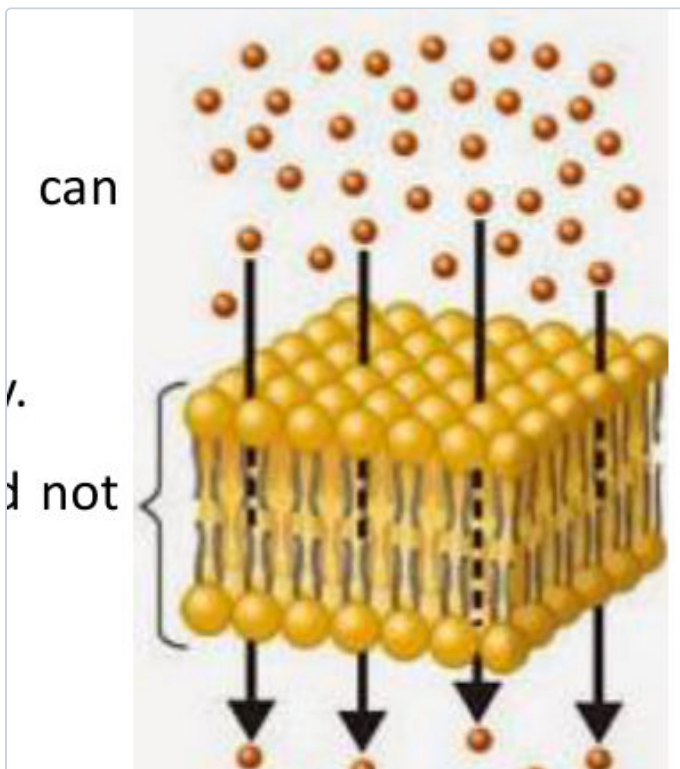
Slide 13: The diffusion equation with the lecturer's notes

[added: this is the basis of Fick's law of diffusion; e.g., thickened alveolar membranes (↑ distance) or lost lung surface (↓ area) reduce gas diffusion.]

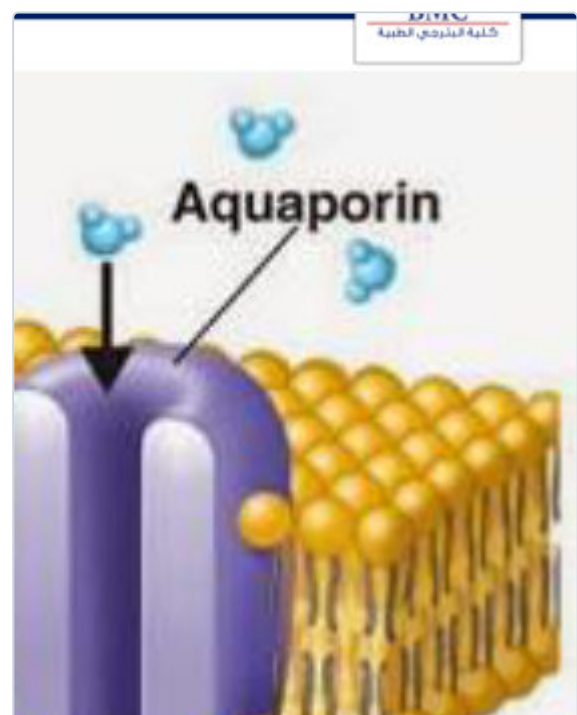
Two routes:

1. Through the lipid bilayer

Substance	How
★ a) Lipid-soluble (O ₂ , N ₂)	Dissolve directly in the bilayer. ★ Rate ∝ lipid solubility . O ₂ enters " almost as though the membrane did not exist "
★ b) Water	Passes directly through the bilayer and through ★ aquaporins . Water is insoluble in lipid but molecules are small with high kinetic energy , so they " penetrate like bullets "
c) Lipid-insoluble molecules	Pass like water if small and uncharged



Slide 14: Lipid-soluble molecules crossing the bilayer



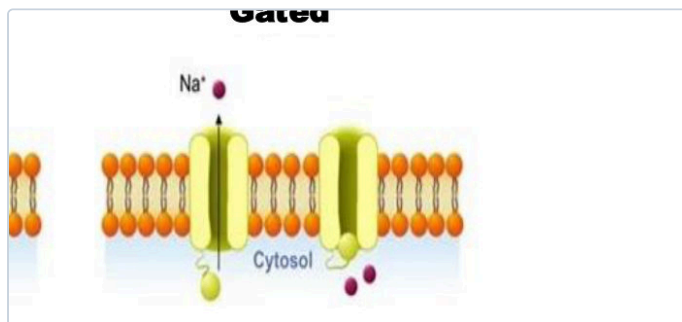
Slide 15: Aquaporin water channel

2. Through protein channels (Slides 16-17)

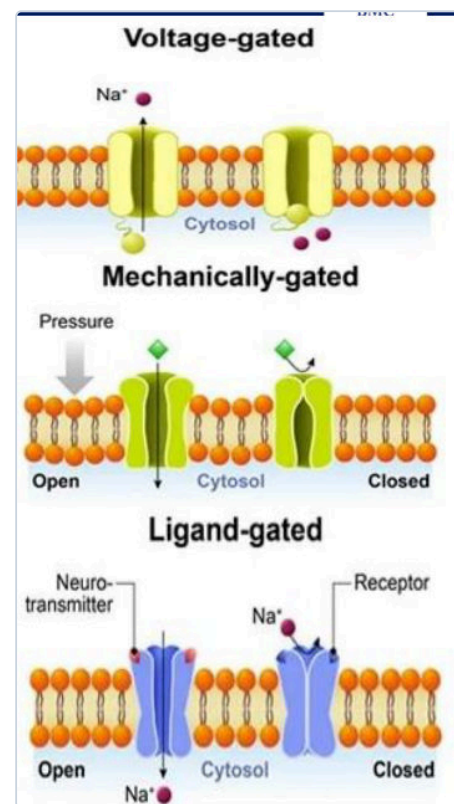
Type	Behaviour
I. Leak channels	★ Always open (non-gated)
II. Gated channels	Gate(s) on the inner, outer, or both surfaces; activity depends on gate opening/closing

★★ Gated channel types:

Gate	Opens/closes by	Example [added]
★ Voltage-gated	Change in transmembrane potential	Na ⁺ channels of the action potential
★ Mechanically gated (stretch/pressure)	Mechanosensitive ; important in cell movement	Touch receptors, hair cells
★ Ligand-gated	Ligand binds a receptor (extracellular or intracellular)	Nicotinic ACh receptor
Phosphorylation-gated	Protein phosphorylation / dephosphorylation	



Slide 16: Leak (always open) vs gated channels



Slide 17: Voltage-, mechanically- and ligand-gated channels

🔄 **Recap: Simple diffusion.** No carrier. Rate rises with ΔC , area and temperature, and falls with distance and $\sqrt{MW}$. Through the bilayer: lipid-soluble gases ($\propto$ solubility), water (bilayer + aquaporins), small uncharged molecules. Through channels: leak (always open) or gated (voltage, mechanical, ligand, phosphorylation).

🔗 **Link:** Large polar molecules like glucose can't use either route, so they need a carrier.

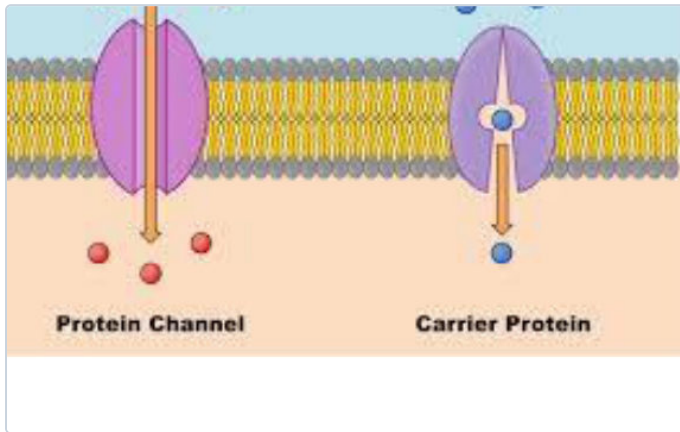
▶ **Up next: Facilitated diffusion.**

2.5 Facilitated (Carrier-Mediated) Diffusion (Slides 18-20)

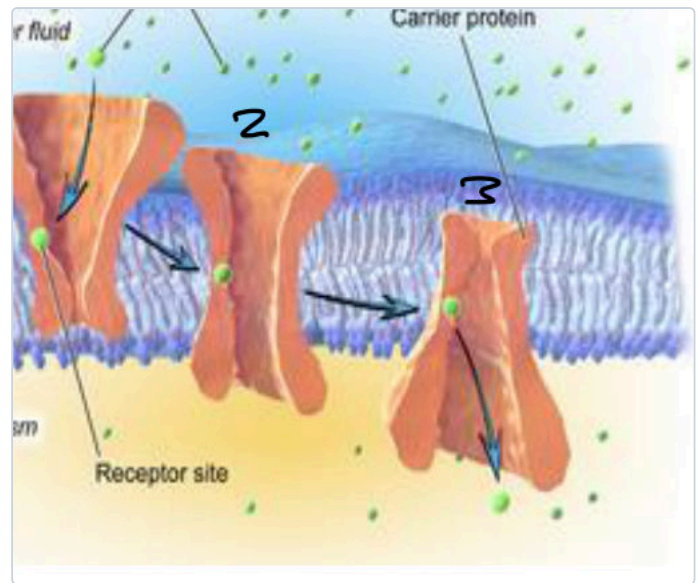
Mechanism:

1. Carrier binds **large molecules** (e.g., ★ **glucose**)

- Carrier has a channel open **part-way** through the membrane
- Molecule enters and **binds a specific receptor** on the carrier
- In a fraction of a second, a ★ **conformational change** opens the channel to the **opposite side**
- Molecule is **released** on the other side



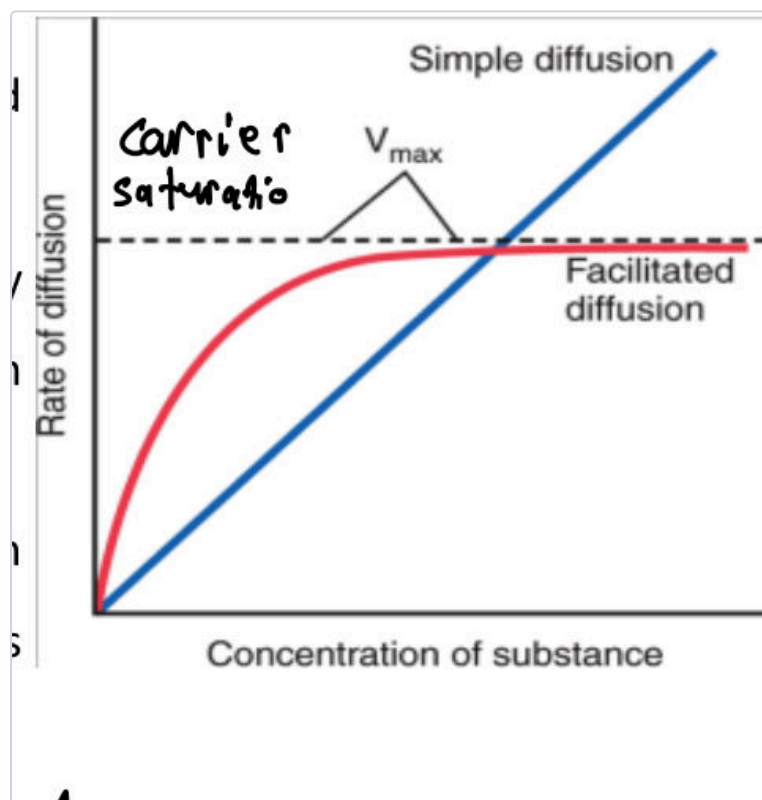
Slide 18: Protein channel vs carrier protein



Slide 19: Carrier conformational change

★★ Properties:

- ★ **Specificity** of each carrier
- ★ **Competition** between similar substances (**glucose vs galactose**) for one carrier
- ★ **Saturation**: rate rises with the gradient **up to a maximum (V_{max})**, limited by how fast the carrier can flip between its two states
- ★ **More temperature-sensitive than simple diffusion** (starred by the lecturer)



Slide 20: Simple diffusion (linear) vs facilitated diffusion (saturates at V_{max})

 **Recap: Facilitated diffusion.** A carrier binds, changes shape and releases on the other side. It is specific, competitive and saturable (V_{max}), and more temperature-sensitive than simple diffusion. It is still passive (down the gradient).

 **Link:** Osmosis, active transport and vesicular transport continue in **Lecture 03**.

 **Up next:** Conclusion and case.

2.6 Conclusion (Slide 22)

- The cell membrane is **amphipathic**
- Proteins **55%**, phospholipids **25%**, cholesterol **13%**, other lipids **4%**, carbohydrates **3%**
- Transport = **diffusion, active, or vesicular**
- Diffusion = **simple, facilitated, or osmosis**
- Simple diffusion = **directly through the membrane** or **through protein channels**

3. High-Yield Comparison Tables

Table A: Simple vs facilitated diffusion

	Simple	Facilitated
Carrier	✗ None	✓ Specific carrier
Energy	None	None
Direction	Down gradient	Down gradient
Rate vs concentration	★ Linear, no maximum	★ Saturates (V_{max})
Specificity / competition	No	★ Yes (glucose vs galactose)
Temperature sensitivity	Less	★ More
Examples	O ₂ , N ₂ , CO ₂ , water, small uncharged molecules, ions via channels	Glucose (GLUT)

Table B: Channel types (see §2.4)

Table C: Integral vs peripheral (per lecturer)

Integral	Peripheral
Channels, facilitated carriers, pumps	Receptors, enzymes

Table D: Junctions

Junction	Function
Tight (zonula occludens)	Seal, fasten cells
Desmosome	Anchor cells (cadherin + keratin filaments on figure)
Zonula adherens	Fasten cells
Gap junction	Communication: ions/molecules pass cell to cell
CAMs	Cell to basal lamina

4. Rules, Exceptions, Patterns

- **Proteins are the biggest component (55%)**, not lipids. Lipids together = 42%.
- **Amphipathic**: hydrophilic phosphate heads outward, hydrophobic fatty-acid tails inward.
- **Diffusion rate $\propto \Delta C \times \text{area} \times T / (\text{distance} \times \sqrt{MW})$** .
- **Lipid-soluble gases cross freely** (rate $\propto$ lipid solubility).
- **Water is lipid-insoluble yet crosses the bilayer** (small, fast), plus aquaporins.
- **Charged ions need channels**; small **uncharged** molecules can use the bilayer.
- **Only facilitated diffusion saturates** among the passive processes.
- **Gap junctions are the only junction for communication**; the others fasten.

5. Clinical Correlations *[added; the lecture has none]*

- **Cystic fibrosis**: defective CFTR, a Cl^- channel (a membrane protein).
- **Diabetes and glucose uptake**: insulin increases GLUT4 carriers in muscle/fat (facilitated diffusion).
- **Pulmonary fibrosis/oedema**: $\uparrow$ diffusion distance $\rightarrow \downarrow \text{O}_2$ transfer (diffusion equation).
- **Nephrogenic diabetes insipidus**: aquaporin-2 defects.
- **Myasthenia gravis**: antibodies against a ligand-gated channel (ACh receptor).
- **Tissue matching / transplant rejection**: identity glycoproteins.

6. Rapid Revision (Last-Day)

- Membrane **7.5-10 nm (75-100 Å)**, elastic, **semipermeable**.
- **Proteins 55 · phospholipids 25 · cholesterol 13 · other lipids 4 · carbohydrates 3**.
- Phospholipid = **clothespin**: polar phosphate head, 2 non-polar fatty-acid tails $\rightarrow$ **amphipathic**.
- Bilayer $\rightarrow$ flexibility + selectivity; **cholesterol $\rightarrow$ permeability + toughness**; carbs $\rightarrow$ recognition, self-differentiation, immunity, adhesion.
- 10 protein functions; channels/carriers/pumps = **integral**, receptors/enzymes = **peripheral** (lecturer); identity = **glycoprotein**.
- Junctions: **tight, desmosome, zonula adherens** fasten; **gap** transfers. CAMs $\rightarrow$ basal lamina.
- Transport: **diffusion** (simple, facilitated, osmosis) · **active** (primary, secondary) · **vesicular** (pino-, phagocytosis).
- **Rate $\propto \Delta C \times \text{area} \times T / (\text{distance} \times \sqrt{MW})$** .
- Bilayer: $\text{O}_2/\text{N}_2 \propto$ lipid solubility; **water like bullets + aquaporins**; small uncharged molecules.
- Channels: **leak (always open)** vs **gated** (voltage, mechanical, ligand, phosphorylation).
- Facilitated: glucose; **specific, competitive, saturable (V_{max}), more temperature-sensitive**.

7. Exam-Style Questions

Q1 (MCQ, lecture case, Slide 23). O_2 moves from alveolar air into RBCs with no carrier or energy; the rate correlates with O_2 's lipid solubility. Mechanism? 1) Active transport via an ATP pump **2) Simple diffusion directly through the lipid bilayer** 3) Ligand-gated channel 4) Facilitated diffusion via a carrier **Answer: 2.** No carrier, no energy, rate $\propto$ lipid solubility = simple diffusion through the bilayer (Slide 14).

Q2 (MCQ). Which property distinguishes facilitated from simple diffusion? a) Requires ATP b) Moves against the gradient **c) Shows a maximum rate (saturation)** d) Occurs only in neurons **Answer: c.** Both are passive and down-gradient; only facilitated has Vmax (plus specificity and competition).

Q3 (MCQ). Doubling the membrane thickness would change the simple diffusion rate how? **Answer: Halve it:** rate is inversely proportional to distance.

Q4 (MCQ). Which junction lets ions pass directly from one cell to the next? a) Tight junction b) Desmosome c) Zonula adherens **d) Gap junction Answer: d.**

Q5 (Short answer). List the membrane composition by percentage and explain "amphipathic".
Answer: Proteins 55%, phospholipids 25%, cholesterol 13%, other lipids 4%, carbohydrates 3%.
Amphipathic = each phospholipid has a hydrophilic polar phosphate head and hydrophobic non-polar fatty-acid tails.

8. Exam Pitfalls & Traps

1. **"Lipids are the main component."** No: **proteins 55%.**
2. **Water can't cross the bilayer.** It can (like bullets) **and** via aquaporins.
3. **Facilitated diffusion uses energy.** No, it's passive; only the carrier is extra.
4. **Simple diffusion saturates.** No; only carrier-mediated processes do.
5. **Rate $\propto$ molecular weight.** It's inversely $\propto \sqrt{MW}$.
6. **Temperature:** facilitated diffusion is **more** temperature-sensitive than simple (starred).
7. **Receptors integral or peripheral?** The lecturer wrote **peripheral** next to receptors and enzymes. Textbooks often call many receptors integral; follow the lecture.
8. **Leak channels have gates.** No, they're always open.
9. **Gap junctions fasten cells.** No, they **communicate**; tight junctions/desmosomes fasten.
10. **⚠ "Zonula occludence / adherence"** on the slide = standard terms **zonula occludens / adherens.**